Indian Premier League IPL - Analysis

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OBJECTIVE QUESTIONS

1. **List the different d types of columns in table "ball_by_ball" (using information schema).**

A. The table `ball_by_ball` consists of **11 columns**, and **each column stores data in the** `INTEGER` **data type**. This ensures that all values in the table represent numeric information, such as IDs, scores, or counts, without any text or decimal data.

```
-- QUERY --
SELECT COLUMN_NAME, DATA_TYPE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Ball_by_Ball';
```

OUTPUT :

	COLUMN_NAME	DATA_TYPE
▶	Match_Id	int
	Over_Id	int
	Ball_Id	int
	Innings_No	int
	Team_Batting	int
	Team_Bowling	int
	Striker_Batting_Position	int
	Striker	int
	Non_Striker	int
	Bowler	int
	Runs_Scored	int

2. **What is the total number of runs scored in 1st season by RCB (bonus: also include the extra runs using the extra runs table).**

A. The **Royal Challengers Bangalore (RCB)** team scored a **total of 2601 runs** (including both batsmen's runs and extra runs) during the **1st season**.

```
-- QUERY --
SELECT  m.Season_Id ,
        SUM(bob.Runs_Scored) + SUM(COALESCE(er.Extra_Runs, 0)) AS Total_Runs_Scored
FROM ball_by_ball bob
JOIN team t ON t.Team_Id = bob.Team_Batting
JOIN matches m ON m.Match_Id = bob.Match_Id
LEFT JOIN extra_runs er ON er.Match_Id = bob.Match_Id
        AND er.Over_Id = bob.Over_Id
        AND er.Ball_Id = bob.Ball_Id
        AND er.Innings_No = bob.Innings_No
WHERE t.Team_Name = 'Royal Challengers Bangalore'
        AND Season_Id = ( SELECT MIN(Season_Id)
                          FROM matches
                          WHERE Match_Id = m.Match_Id )
GROUP BY m.Season_Id
ORDER BY m.Season_Id  LIMIT 1 ;
```

OUTPUT :

	Season_Id	Total_Runs_Scored
▶	6	2601

3. How many players were more than the age of 25 during season 2014?

A. During the 2014 IPL season, a total of 89 players were above 25 years of age.

```
-- QUERY --
SELECT COUNT(DISTINCT p.Player_Id) AS Players_Count
FROM player p
JOIN player_match pm ON pm.Player_Id = p.Player_Id
JOIN matches m ON m.Match_Id = pm.Match_Id
WHERE YEAR(m.Match_Date) = 2014
    AND TIMESTAMPDIFF(YEAR, p.DOB , '2014-01-01') > 25 ;
```

OUTPUT :

	Players_Count
▶	89

4. How many matches did RCB win in 2013?

A. The Royal Challengers Bangalore (RCB) team won a total of 9 matches during the 2013 IPL season.

```
-- QUERY --
SELECT COUNT(*) AS Matches_Won
FROM team t
JOIN matches m ON t.Team_Id = m.Team_1 OR t.Team_Id = m.Team_2
WHERE Match_Winner = 2
    AND Team_Name = 'Royal Challengers Bangalore'
    AND YEAR(Match_Date) = 2013;
```

OUTPUT :

	Matches_Won
▶	9

5. List the top 10 players according to their strike rate in the last 4 seasons.

A. The top 10 players with the highest strike rates in the last 4 IPL seasons are identified. Strike rate is a performance metric calculated as the (total runs scored / total balls faced) × 100, representing how quickly a player scores runs.

```
-- QUERY --
WITH Prev_Season AS (
    SELECT DISTINCT Season_Year , Season_Id
    FROM season
    ORDER BY Season_Year DESC LIMIT 4
    )
SELECT DISTINCT p.Player_Name ,
        ( SUM(bob.Runs_Scored) * 100 / COUNT(bob.Ball_Id) ) AS Strike_Rate
FROM ball_by_ball bob
JOIN player p ON bob.Striker = p.Player_Id
JOIN matches m ON bob.Match_Id = m.Match_Id
JOIN Prev_Season ps ON m.Season_Id = ps.Season_Id
GROUP BY p.Player_Name
ORDER BY Strike_Rate DESC LIMIT 10 ;
```

Player_Name	Strike_Rate
RV Gomez	400.0000
AN Ahmed	233.3333
CR Brathwaite	207.5000
PJ Sangwan	200.0000
Bipul Sharma	192.5926
KH Pandya	186.6142
B Lee	175.0000
M Morkel	173.6842
SN Khan	171.8447
BCJ Cutting	169.7674

6. What are the average runs scored by each batsman considering all the seasons?

A. The average runs scored by each batsman have been calculated across all IPL seasons.
This provides insight into each player's overall batting consistency and performance throughout the tournament.
The calculation involves grouping total runs by player and dividing by the number of matches they have played.

```
-- QUERY --
WITH CTE AS (
    SELECT p.Player_Id , p.Player_Name ,
        SUM(bob.Runs_Scored) AS Total_Runs ,
        COUNT(DISTINCT bob.Match_Id) AS Total_Innings
    FROM ball_by_ball bob
    JOIN player p ON bob.Striker = p.Player_Id
    GROUP BY  p.Player_Id , p.Player_Name )
SELECT CTE.* ,
    ROUND(Total_Runs/Total_Innings , 2) AS Average_Runs
FROM CTE
ORDER BY Average_Runs DESC ;
```

Player_Id	Player_Name	Total_Runs	Total_Innings	Average_Runs
383	LMP Simmons	942	22	42.82
8	V Kohli	2472	62	39.87
187	DA Warner	2348	61	38.49
431	N Rana	104	3	34.67
110	AB de Villiers	1968	57	34.53
19	MEK Hussey	1019	30	33.97
162	CH Gayle	1634	49	33.35
85	AM Rahane	1847	57	32.40
100	SE Marsh	622	20	31.10
46	RV Uthappa	1852	60	30.87
147	DR Smith	1707	57	29.95
57	RG Sharma	1899	64	29.67
413	KH Pandya	237	8	29.63

7. What are the average wickets taken by each bowler considering all the seasons?

A. The average wickets taken by each bowler have been calculated across all IPL seasons. The calculation involves summing up the total wickets taken by each bowler and dividing it by the number of matches they bowled in.

```
-- QUERY --
WITH CTE AS (
    SELECT bob.Bowler , p.Player_Name ,
        COUNT(wt.Player_Out) AS Total_Wickets ,
        COUNT(DISTINCT bob.Match_Id) AS Innings_Played
    FROM ball_by_ball bob
    JOIN player p ON bob.Bowler = p.Player_Id
    JOIN wicket_taken wt ON  bob.Match_Id = wt.Match_Id
                    AND bob.Over_Id = wt.Over_Id
                    AND bob.Ball_Id = wt.Ball_Id
                    AND bob.Innings_No = wt.Innings_No
    GROUP BY bob.Bowler, p.Player_Name )
SELECT CTE.* ,
    ROUND(Total_Wickets / Innings_Played ,2) AS Average_Wickets
FROM CTE
ORDER BY Total_Wickets DESC ;
```

Bowler	Player_Name	Total_Wickets	Innings_Played	Average_Wickets
71	DJ Bravo	81	42	1.93
364	MM Sharma	78	42	1.86
299	B Kumar	77	43	1.79
315	SP Narine	68	37	1.84
50	Harbhajan Singh	67	42	1.60
194	SL Malinga	66	36	1.83
310	JP Faulkner	61	32	1.91
367	Sandeep Sharma	59	34	1.74
382	YS Chahal	58	34	1.71
73	A Nehra	57	29	1.97
35	RA Jadeja	56	29	1.93
232	UT Yadav	56	28	2.00
136	A Mishra	55	32	1.72

8. List all the players who have average runs scored greater than the overall average and who have taken wickets greater than the overall average.

A. The goal is to identify players who have both batted and bowled exceptionally well - Specifically , those who have: An average runs scored greater than the overall average runs of all players, and Wickets taken greater than the overall average wickets of all players.

```
-- QUERY--
WITH batting AS (
    SELECT
        p.Player_ID,
        p.Player_Name,
        COALESCE(SUM(b.Runs_Scored), 0) AS total_runs,
        COUNT(w.Player_Out) AS times_out,
        CASE
            WHEN COUNT(w.Player_Out) = 0 THEN COALESCE(SUM(b.Runs_Scored), 0)
            ELSE CAST(SUM(b.Runs_Scored) AS FLOAT) / COUNT(w.Player_Out)
        END AS batting_average
    FROM player p
    LEFT JOIN ball_by_ball b
        ON p.Player_ID = b.Striker
    LEFT JOIN wicket_taken w
        ON w.Player_Out = p.Player_ID
           AND w.Match_Id = b.Match_Id
           AND w.Innings_No = b.Innings_No
           AND w.Over_Id = b.Over_Id
           AND w.Ball_Id = b.Ball_Id
    GROUP BY p.Player_ID, p.Player_Name
),
```

```sql
bowling AS (
    SELECT
        p.Player_ID,
        p.Player_Name,
        COUNT(w.Player_Out) AS wickets,
        COALESCE(SUM(b.Runs_Scored),0) AS runs_conceded,
        CASE
            WHEN COUNT(w.Player_Out) = 0 THEN NULL
            ELSE CAST(SUM(b.Runs_Scored) AS FLOAT) / COUNT(w.Player_Out)
        END AS bowling_average
    FROM player p
    LEFT JOIN ball_by_ball b
        ON p.Player_ID = b.Bowler
    LEFT JOIN wicket_taken w
        ON w.Match_Id = b.Match_Id
           AND w.Innings_No = b.Innings_No
           AND w.Over_Id = b.Over_Id
           AND w.Ball_Id = b.Ball_Id
    GROUP BY p.Player_ID, p.Player_Name
),
```

```sql
overall AS (
    -- Overall Batting Average
    SELECT
        CAST(SUM(b.Runs_Scored) AS FLOAT) / NULLIF(SUM(
            CASE WHEN w.Player_Out IS NOT NULL THEN 1 ELSE 0 END
        ),0) AS overall_batting_avg,
        -- Overall Wickets Average per player
        (SELECT AVG(wickets_count)
         FROM (
            SELECT COUNT(*) AS wickets_count
            FROM wicket_taken
            GROUP BY Player_Out
         ) x) AS overall_wickets_avg
    FROM ball_by_ball b
    LEFT JOIN wicket_taken w
        ON b.Ball_Id = w.Ball_Id
)
```

```sql
SELECT
    ba.Player_ID,
    ba.Player_Name,
    ba.total_runs AS runs,
    ROUND(ba.batting_average,2) AS batting_average,
    bo.wickets,
    ROUND(bo.bowling_average,2) AS bowling_average
FROM batting ba
JOIN bowling bo
    ON ba.Player_ID = bo.Player_ID
CROSS JOIN overall o
WHERE ba.batting_average > o.overall_batting_avg
  AND bo.wickets > o.overall_wickets_avg
  AND ba.total_runs > 300 AND bo.wickets > 10
ORDER BY Player_ID ;
```

OUTPUT :

Player_ID	Player_Name	runs	batting_average	wickets	bowling_average
9	JH Kallis	462	23.1	23	24.65
27	Yuvraj Singh	1098	24.95	13	35.08
32	SR Watson	1315	27.4	54	21.52
35	RA Jadeja	670	23.93	56	24.8
50	Harbhajan Singh	353	17.65	67	24.09
71	DJ Bravo	423	17.62	81	17
147	DR Smith	1707	31.61	15	21
154	JP Duminy	1015	46.14	13	31.15
163	MC Henriques	643	25.72	37	22.51
191	AD Mathews	316	24.31	13	36.23

9. Create a table rcb_record table that shows the wins and losses of RCB in an individual venue.

A. To obtain RCB's record, I calculated the number of matches won and lost by RCB at each venue.

```
-- QUERY --
CREATE TABLE rcb_record AS
SELECT v.Venue_Name ,
        SUM(CASE WHEN (m.Team_1 = 2 OR m.Team_2 = 2)
                AND m.Match_Winner = 2 THEN 1 ELSE 0 END ) AS Wins ,
        SUM(CASE WHEN (m.Team_1 = 2 OR m.Team_2 = 2)
                AND m.Match_Winner != 2 AND m.Match_Winner IS NOT NULL
                THEN 1 ELSE 0 END ) AS Losses
FROM matches m
JOIN venue v ON m.Venue_Id = v.Venue_Id
GROUP BY v.Venue_Name ;
SELECT *
FROM rcb_record ;
```

OUTPUT :

	Venue_Name	Wins	Losses
▶	Eden Gardens	2	1
	M Chinnaswamy Stadium	16	11
	Rajiv Gandhi International Stadium, Uppal	1	3
	Feroz Shah Kotla	2	0
	MA Chidambaram Stadium, Chepauk	0	2
	Subrata Roy Sahara Stadium	1	0
	Sawai Mansingh Stadium	0	1
	Wankhede Stadium	1	3
	Punjab Cricket Association Stadium, Mohali	0	2
	Himachal Pradesh Cricket Association Stadium	0	0
	Shaheed Veer Narayan Singh International S...	1	0
	JSCA International Stadium Complex	1	2

10. What is the impact of bowling style on wickets taken?

A. By analyzing the wickets taken by each bowling style, I found that the *Right Arm Medium* bowlers have taken the highest number of wickets. The SQL query used for this analysis is shown below.

```
-- QUERY --
SELECT bs.Bowling_skill ,
       COUNT(*) AS Total_Wickets
FROM bowling_style bs
JOIN player p ON bs.Bowling_Id = p.Bowling_skill
JOIN ball_by_ball bob ON bob.Bowler = p.Player_Id
JOIN wicket_taken wt ON  bob.Match_Id = wt.Match_Id
                    AND bob.Over_Id = wt.Over_Id
                    AND bob.Ball_Id = wt.Ball_Id
                    AND bob.Innings_No = wt.Innings_No
WHERE bs.Bowling_skill IS NOT NULL
GROUP BY bs.Bowling_skill
ORDER BY Total_Wickets DESC ;
```

OUTPUT :

	Bowling_skill	Total_Wickets
▶	Right-arm medium	617
	Right-arm fast-medium	382
	Right-arm medium-fast	343
	Right-arm offbreak	290
	Slow left-arm orthodox	283
	Right-arm fast	237
	Left-arm fast-medium	186
	Legbreak googly	167
	Legbreak	152
	Left-arm medium-fast	114
	Left-arm fast	99
	Left-arm medium	42
	Slow left-arm chinaman	31

11. Write the SQL query to provide a status of whether the performance of the team is better than the previous year's performance on the basis of the number of runs scored by the team in the season and the number of wickets taken.

A. To compare each team's performance with the previous year based on total runs scored and wickets taken, I wrote a query that provides a status indicating whether the team performed better or not compared to the last season. The query used to achieve this is shown below.

```
-- QUERY --
WITH team_runs AS (
    SELECT t.Team_Name, s.Season_Year,
            SUM(b.Runs_Scored) AS Total_Runs
    FROM team t
    JOIN matches m ON t.Team_Id = m.Team_1 OR t.Team_Id = m.Team_2
    JOIN season s ON m.Season_Id = s.Season_Id
    JOIN ball_by_ball b ON m.Match_Id = b.Match_Id
    WHERE b.Team_Batting = t.Team_Id
    GROUP BY t.Team_Name, s.Season_Year
),
```

```sql
team_wickets AS (
        SELECT t.Team_Name, s.Season_Year,
                COUNT(w.Player_Out) AS Total_Wickets
        FROM team t
        JOIN matches m ON t.Team_Id = m.Team_1 OR t.Team_Id = m.Team_2
        JOIN season s ON m.Season_Id = s.Season_Id
        JOIN ball_by_ball b ON m.Match_Id = b.Match_Id
        JOIN wicket_taken w ON b.Match_Id = w.Match_Id
            AND b.Over_Id = w.Over_Id
            AND b.Ball_Id = w.Ball_Id
            AND b.Innings_No = w.Innings_No
        WHERE b.Team_Bowling = t.Team_Id
        GROUP BY t.Team_Name, s.Season_Year
),
Wickets_Runs AS (
        SELECT r.Team_Name, r.Season_Year, r.Total_Runs,
            LAG(r.Total_Runs) OVER (PARTITION BY r.Team_Name ORDER BY r.Season_Year) AS Prev_Runs,
            COALESCE(w.Total_Wickets, 0) AS Total_Wickets,
            LAG(w.Total_Wickets) OVER (PARTITION BY w.Team_Name ORDER BY w.Season_Year) AS Prev_Wickets
        FROM team_runs r
        LEFT JOIN team_wickets w ON r.Team_Name = w.Team_Name
                AND r.Season_Year = w.Season_Year
)
SELECT Team_Name, Season_Year, Total_Runs, Prev_Runs, Total_Wickets, Prev_Wickets,
        CASE
            WHEN Prev_Runs IS NULL OR Prev_Wickets IS NULL THEN "No Previous Data"
            WHEN Total_Runs > Prev_Runs AND Total_Wickets > Prev_Wickets THEN "Increased"
            WHEN Total_Runs < Prev_Runs AND Total_Wickets < Prev_Wickets THEN "Decreased"
            ELSE "Mixed"
        END AS Performance_Status
FROM Wickets_Runs
ORDER BY Team_Name, Season_Year;
```

OUTPUT :

Team_Name	Season_Year	Total_Runs	Prev_Runs	Total_Wickets	Prev_Wickets	Performance_Status
Chennai Super Kings	2013	2671	NULL	123	NULL	No Previous Data
Chennai Super Kings	2014	2519	2671	98	123	Decreased
Chennai Super Kings	2015	2564	2519	113	98	Increased
Delhi Daredevils	2013	2114	NULL	76	NULL	No Previous Data
Delhi Daredevils	2014	1978	2114	60	76	Decreased
Delhi Daredevils	2015	2061	1978	74	60	Increased
Delhi Daredevils	2016	1995	2061	77	74	Mixed
Gujarat Lions	2016	2318	NULL	81	NULL	No Previous Data
Kings XI Punjab	2013	2312	NULL	82	NULL	No Previous Data
Kings XI Punjab	2014	2824	2312	113	82	Increased
Kings XI Punjab	2015	1902	2824	72	113	Decreased
Kings XI Punjab	2016	2051	1902	73	72	Increased
Kolkata Knight Riders	2013	2134	NULL	97	NULL	No Previous Data
Kolkata Knight Riders	2014	2382	2134	95	97	Mixed

12. Can you derive more KPIs for the team strategy?

A. Some advanced key performance indicators (KPIs) for team strategy include:

1. Powerplay Performance

```
WITH powerplay_runs AS (
    SELECT
        m.Season_Id,
        b.Team_Batting,
        SUM(b.Runs_Scored) AS Total_Runs
    FROM ball_by_ball b
    JOIN matches m ON b.Match_Id = m.Match_Id
    WHERE b.Over_Id IN (1, 2, 3, 4,5,6)
    GROUP BY m.Season_Id, b.Team_Batting )
SELECT
    t.Team_Name,
    pr.Season_Id,
    pr.Total_Runs
FROM powerplay_runs pr
JOIN team t ON pr.Team_Batting = t.Team_Id
WHERE Team_Name = 'Royal Challengers Bangalore'
ORDER BY pr.Season_Id;
```

OUTPUT :

Team_Name	Season_Id	Total_Runs
Royal Challengers Bangalore	6	686
Royal Challengers Bangalore	7	493
Royal Challengers Bangalore	8	688
Royal Challengers Bangalore	9	739

2. Batting Performance

```
SELECT
    p.Player_Name,
    SUM(b.Runs_Scored) AS Total_Runs,
    ROUND(SUM(b.Runs_Scored) * 100.0 / COUNT(b.Ball_Id), 2) AS Strike_Rate
FROM ball_by_ball b
JOIN player p ON b.Striker = p.Player_Id
WHERE Striker_Batting_Position BETWEEN 1 AND 8
GROUP BY p.Player_Name
ORDER BY Total_Runs DESC, Strike_Rate DESC, p.Player_Name;
```

OUTPUT :

	Player_Name	Total_Runs	Strike_Rate
▶	V Kohli	2472	135.68
	DA Warner	2348	140.94
	AB de Villiers	1968	164.27
	RG Sharma	1899	132.52
	RV Uthappa	1852	127.11
	AM Rahane	1847	116.90
	SK Raina	1844	133.62
	DR Smith	1707	127.10
	CH Gayle	1634	137.43

3. Bowling Performance

```
WITH bowling_stats AS (
    SELECT
        b.Bowler,
        SUM(b.Runs_Scored) AS Runs_Conceded,
        COUNT(*) AS Balls_Bowled
    FROM ball_by_ball b
    GROUP BY b.Bowler),
wickets_by_bowler AS (
    SELECT
        b.Bowler,
        COUNT(*) AS Wickets
    FROM ball_by_ball b
    JOIN wicket_taken w
        ON b.Match_Id = w.Match_Id
        AND b.Innings_No = w.Innings_No
        AND b.Over_Id = w.Over_Id
        AND b.Ball_Id = w.Ball_Id
    WHERE w.Player_Out IS NOT NULL
    GROUP BY b.Bowler)
SELECT
    p.Player_Name,
    COALESCE(w.Wickets, 0) AS Wickets,
    ROUND(bs.Runs_Conceded * 6.0 / bs.Balls_Bowled, 2) AS Economy
FROM bowling_stats bs
JOIN player p ON bs.Bowler = p.Player_Id
LEFT JOIN wickets_by_bowler w ON bs.Bowler = w.Bowler
ORDER BY Wickets DESC, Economy ASC;
```

OUTPUT :

	Player_Name	Wickets	Economy
▶	DJ Bravo	81	7.77
	MM Sharma	78	7.62
	B Kumar	77	6.53
	SP Narine	68	6.24
	Harbhajan Singh	67	6.84
	SL Malinga	66	6.58
	JP Faulkner	61	7.89
	Sandeep Sharma	59	7.11
	YS Chahal	58	7.66
	A Nehra	57	7.49
	RA Jadeja	56	7.58
	UT Yadav	56	8.04

4. Dot Ball Percentage (Batting)

```
SELECT
  p.Player_Name,
  ROUND(SUM(CASE WHEN b.Runs_Scored = 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS Dot_Ball_Percentage
FROM ball_by_ball b
JOIN player p ON b.Striker = p.Player_ID
WHERE b.Team_Batting = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
GROUP BY p.Player_Name;
```

OUTPUT :

	Player_Name	Dot_Ball_Percentage
▶	CH Gayle	46.17
	TM Dilshan	59.79
	V Kohli	32.11
	MA Agarwal	51.56
	DT Christian	68.75
	KK Nair	66.67
	KB Arun Karthik	45.00
	MC Henriques	36.22
	R Vinay Kumar	34.09
	M Kartik	33.33
	JD Unadkat	0.00
	AB de Villiers	27.55

5. Dot Ball Percentage (Bowling)

```
SELECT
  p.Player_Name,
  ROUND(SUM(CASE WHEN b.Runs_Scored = 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS Dot_Ball_Percentage
FROM ball_by_ball b
JOIN player p ON b.Bowler = p.Player_ID
WHERE b.Team_Bowling = (SELECT Team_ID FROM team WHERE Team_Name = 'Royal Challengers Bangalore')
GROUP BY p.Player_Name;
```

OUTPUT :

Player_Name	Dot_Ball_Percentage
JD Unadkat	36.12
DT Christian	42.11
R Vinay Kumar	34.75
M Muralitharan	37.92
M Kartik	31.95
MC Henriques	40.91
TM Dilshan	0.00
RP Singh	36.25
R Rampaul	50.36
J Syed Mohammad	19.35
AB McDonald	33.33

6. Phase-Wise Performance

```
SELECT
  p.Player_Name,
  CASE
    WHEN b.Over_Id BETWEEN 1 AND 6 THEN 'Powerplay'
    WHEN b.Over_Id BETWEEN 7 AND 15 THEN 'Middle Overs'
    ELSE 'Death Overs'
  END AS Phase,
  COUNT(*) AS Balls_Faced,
  SUM(b.Runs_Scored) AS Runs,
  ROUND(SUM(b.Runs_Scored * 1.0) / COUNT(*), 2) AS Runs_Per_Ball
FROM ball_by_ball b
JOIN player p ON b.Striker = p.Player_ID
WHERE b.Team_Batting = (SELECT Team_ID
                        FROM team WHERE
                        Team_Name = 'Royal Challengers Bangalore')
GROUP BY p.Player_Name, Phase;
```

OUTPUT :

Player_Name	Phase	Balls_Faced	Runs	Runs_Per_Ball
CH Gayle	Powerplay	712	886	1.24
TM Dilshan	Powerplay	79	53	0.67
V Kohli	Powerplay	607	681	1.12
MA Agarwal	Powerplay	48	49	1.02
MA Agarwal	Middle Overs	11	14	1.27
DT Christian	Middle Overs	14	4	0.29
CH Gayle	Middle Overs	383	560	1.46
KK Nair	Middle Overs	7	1	0.14
KB Arun Karthik	Middle Overs	15	19	1.27
CH Gayle	Death Overs	94	188	2.00
KB Arun Karthik	Death Overs	25	22	0.88
KK Nair	Powerplay	5	8	1.60
V Kohli	Middle Overs	965	1302	1.35

13. Using SQL, write a query to find out the average wickets taken by each bowler in each venue. Also, rank them according to the average value.

A. Using SQL, I calculated the average number of wickets taken by each bowler at every venue and ranked them based on their average wickets. The query used to obtain this result is given below.

```
-- QUERY --
WITH COMBINED AS (
        SELECT v.Venue_Name , p.Player_Name ,
                ROUND(COUNT(wt.Player_Out) * 1.0 / COUNT(DISTINCT bob.Match_Id) , 2) AS Avg_Wickets_Taken
        FROM ball_by_ball bob
        JOIN player p ON bob.Bowler = p.Player_Id
        JOIN wicket_taken wt ON bob.Match_Id = wt.Match_Id
                                AND bob.Over_Id = wt.Over_Id
                                AND bob.Ball_Id = wt.Ball_Id
                                AND bob.Innings_No = wt.Innings_No
        JOIN matches m ON m.Match_Id = bob.Match_Id
        JOIN venue v ON v.Venue_Id = m.Venue_Id
        GROUP BY v.Venue_Name , p.Player_Name ) ,
RANKED AS (
        SELECT Venue_Name , Player_Name , Avg_Wickets_Taken ,
                DENSE_RANK() OVER(ORDER BY Avg_Wickets_Taken DESC) AS Rnk
        FROM COMBINED )
SELECT *
FROM RANKED
ORDER BY  Rnk ;
```

OUTPUT :

	Venue_Name	Player_Name	Avg_Wickets_Taken	Rnk
▶	Punjab Cricket Association Stadium, Mohali	DJG Sammy	6.00	1
	Feroz Shah Kotla	JD Unadkat	5.00	2
	Rajiv Gandhi International Stadium, Uppal	JP Faulkner	5.00	2
	Sheikh Zayed Stadium	JP Faulkner	5.00	2
	Brabourne Stadium	CH Morris	4.00	3
	Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket ...	A Nehra	4.00	3
	Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket ...	JP Duminy	4.00	3
	Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket ...	MP Stoinis	4.00	3
	Dubai International Cricket Stadium	RA Jadeja	4.00	3
	Feroz Shah Kotla	AD Russell	4.00	3
	Himachal Pradesh Cricket Association Stadium	SL Malinga	4.00	3
	JSCA International Stadium Complex	SP Narine	4.00	3
	M Chinnaswamy Stadium	Karanveer S...	4.00	3
	MA Chidambaram Stadium, Chepauk	JP Faulkner	4.00	3

14. Which of the given players have consistently performed well in past seasons? (Will you use any visualization to solve the problem?)

A. To identify players who have consistently performed well across multiple seasons, I analyzed both batsmen and bowlers based on their total innings played and cumulative runs or wickets. Visualizations such as bar charts or line graphs can be used to compare consistency across seasons.

1. **Consistent Performance of the batsman by Innings Played and Total Runs Scored in the seasons.**

   Consistent performance across seasons indicates reliability and adaptability under different conditions. Players with high total runs and steady averages over many innings are considered consistent performers.

   Approach :

   - Aggregate player data by season using total runs, innings, and averages.
   - Visualize trends using line charts to track performance consistency across seasons.
   - Compare standard deviation of runs/averages to measure fluctuations in performance.
   -

   INSIGHTS :

   - V. Kohli and D.A. Warner show high total runs with stable averages, indicating consistent top-tier performance.
   - Players like A.B. de Villiers and R.G. Sharma maintain strong averages but with slightly fewer innings.

- Lower variance in averages across seasons suggests dependable form and adaptability to conditions.

RECOMMENDATIONS :

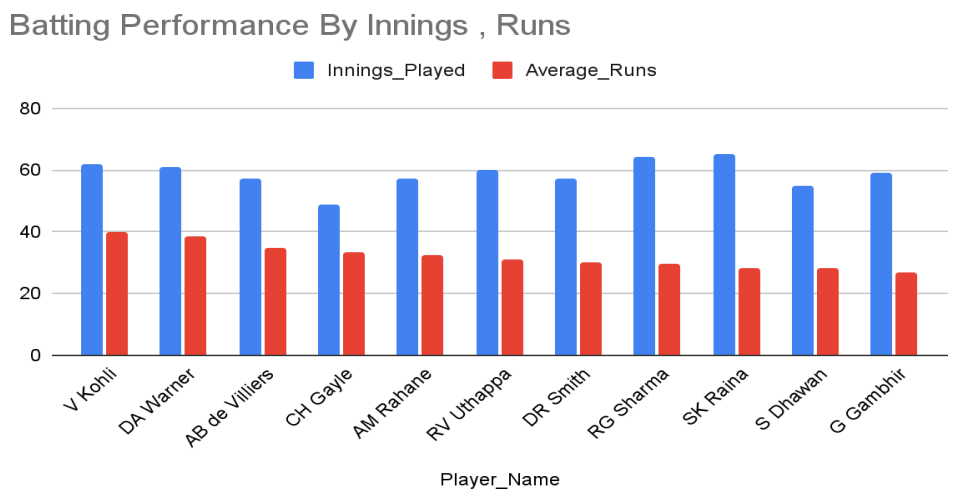
- Retain and build strategies around consistently high performers like Kohli and Warner.
- Monitor players with fluctuating performances for targeted training and workload management.
- Use seasonal trend visualizations regularly to identify emerging consistent performers early.

```sql
WITH cte AS (
    SELECT DISTINCT b.Striker AS Player_ID, p.Player_Name,
        SUM(b.Runs_Scored) AS Total_Runs,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        ROUND((SUM(b.Runs_Scored) / COUNT(DISTINCT b.Match_Id)), 2) AS Avg_Runs
    FROM ball_by_ball b
    JOIN player p ON b.Striker = p.Player_Id
    JOIN matches m ON b.Match_Id = m.Match_Id
    GROUP BY b.Striker, p.Player_Name
)
SELECT *
FROM cte
WHERE Total_Runs > 1500 AND Innings_Played > 40
ORDER BY Total_Runs DESC, Innings_Played DESC;
```

OUTPUT :

Player_ID	Player_Name	Total_Runs	Innings_Played	Avg_Runs
8	V Kohli	2472	62	39.87
187	DA Warner	2348	61	38.49
110	AB de Villiers	1968	57	34.53
57	RG Sharma	1899	64	29.67
46	RV Uthappa	1852	60	30.87
85	AM Rahane	1847	57	32.40
21	SK Raina	1844	65	28.37
147	DR Smith	1707	57	29.95
162	CH Gayle	1634	49	33.35
40	G Gambhir	1569	59	26.59
42	S Dhawan	1542	55	28.04

VISUALIZATION :



2. **Consistent Performance of the bowlers by Innings Played and Total Wickets Taken in the seasons.**

   Consistent bowlers maintain strong wicket-taking ability across multiple seasons while sustaining performance over many innings. Analyzing wickets and innings played helps identify reliability and long-term impact.

   APPROACH :

   - Aggregate data for each bowler using total wickets and innings played across all seasons.
   - Use bar charts or line graphs to visualize consistency in wickets taken per season.
   - Compare average wickets per innings to assess steady performers versus occasional peaks.

   INSIGHTS :

   - DJ Bravo, MM Sharma, and B Kumar show top consistency with high wickets and many innings.
   - SP Narine and SL Malinga display excellent averages despite fewer innings, indicating efficiency.
   - Low fluctuation in average wickets shows sustained impact across multiple seasons.

   RECOMMENDATIONS :

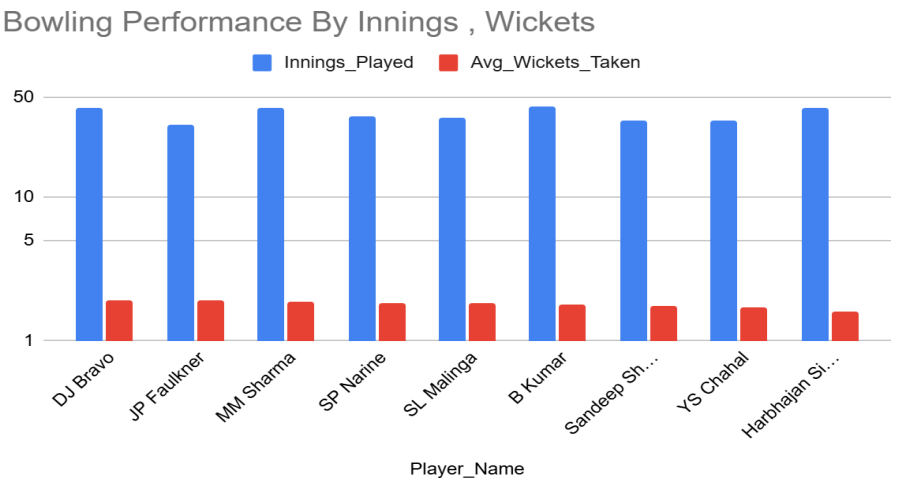
   - Retain core bowlers like Bravo and Sharma for crucial matches due to their proven consistency.
   - Provide injury and workload management to sustain long-term performance of key bowlers.
   - Use trend visualization each season to detect rising consistent wicket-takers early.

```sql
WITH cte AS (
    SELECT DISTINCT b.Bowler AS Bowler_Id, p.Player_Name,
        COUNT(w.Player_Out) AS Total_Wickets,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        ROUND((COUNT(w.Player_Out) / COUNT(DISTINCT b.Match_Id)), 2) AS Avg_Wickets_Taken
    FROM ball_by_ball b
    JOIN player p ON b.Bowler = p.Player_Id
    JOIN wicket_taken w ON (b.Match_Id = w.Match_Id
        AND b.Over_Id = w.Over_Id
        AND b.Ball_Id = w.Ball_Id
        AND b.Innings_No = w.Innings_No)
    JOIN matches m ON b.Match_Id = m.Match_Id
    GROUP BY b.Bowler, p.Player_Name
)
SELECT *
FROM cte
WHERE Total_Wickets > 55 AND Innings_Played > 30
ORDER BY Total_Wickets DESC, Innings_Played DESC;
```

OUTPUT :

Bowler_Id	Player_Name	Total_Wickets	Innings_Played	Avg_Wickets_Taken
71	DJ Bravo	81	42	1.93
364	MM Sharma	78	42	1.86
299	B Kumar	77	43	1.79
315	SP Narine	68	37	1.84
50	Harbhajan Singh	67	42	1.60
194	SL Malinga	66	36	1.83
310	JP Faulkner	61	32	1.91
367	Sandeep Sharma	59	34	1.74
382	YS Chahal	58	34	1.71

VISUALIZATION :



15. Are there players whose performance is more suited to specific venues or conditions? (How would you present this using charts?)

A. To find players whose performance is more suited to specific venues, I analyzed both batsmen and bowlers by comparing their average runs or wickets at each venue with their overall averages. The results can be effectively presented using bar charts or heat maps to visualize performance variations across different stadiums.

1. **Batsman whose performance are more suited to specific Venues:**

Player performance in cricket can vary significantly depending on the venue due to unique pitch conditions and local factors. Identifying players who excel at specific venues helps optimize team strategies and selection.

APPROACH :

- Calculate player-wise average performance at each venue and compare it to their overall average.
- Use SQL or spreadsheet analysis to filter for players with a minimum number of innings at each venue to ensure meaningful data.
- Visualize the difference using bar charts for individuals or heat maps for broader trends.

INSIGHTS :

- Certain players show significantly higher average runs at specific stadiums, highlighting venue preference or adaptability.
- Some players perform better than their career average in a few locations, indicating a potential "home advantage."
- Visual charts make it easy to identify these outlier performances quickly.

RECOMMENDATIONS :

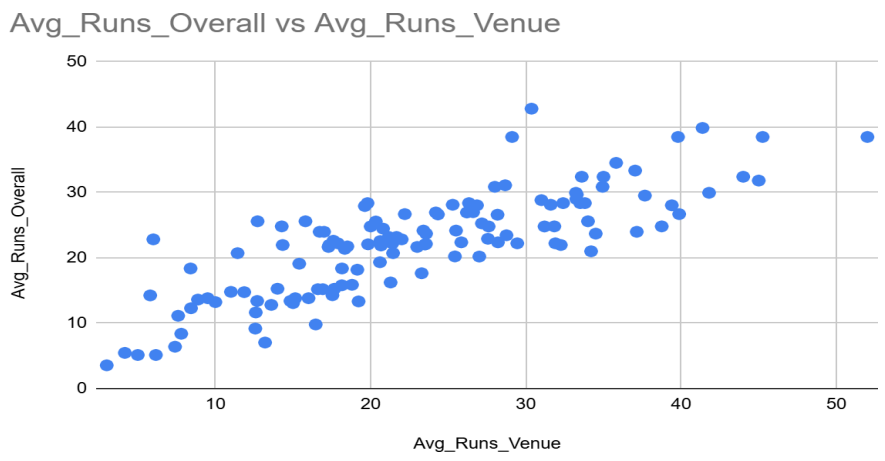
- Prioritize selecting players for matches at venues where they have historically outperformed their average.
- Use bar charts to compare individual players' venue-specific averages against their overall career figures.
- Implement heat maps for team management to strategize broader combinations based on venue-specific strengths

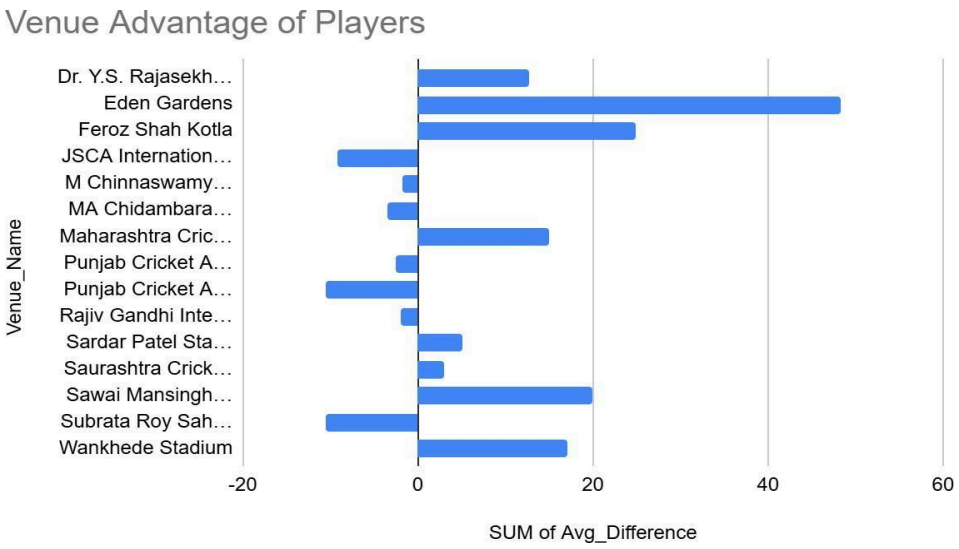
```sql
WITH player_venue_stats AS (
    SELECT p.Player_Id, p.Player_Name, v.Venue_Id, v.Venue_Name,
        SUM(b.Runs_Scored) AS Total_Runs,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        SUM(b.Runs_Scored) * 1.0 / COUNT(DISTINCT b.Match_Id) AS Avg_Runs_Venue
    FROM ball_by_ball b
    JOIN player p ON b.Striker = p.Player_Id
    JOIN matches m ON b.Match_Id = m.Match_Id
    JOIN venue v ON m.Venue_Id = v.Venue_Id
    GROUP BY p.Player_Id, p.Player_Name, v.Venue_Id, v.Venue_Name
),
player_overall_stats AS (
    SELECT p.Player_Id, p.Player_Name,
        SUM(b.Runs_Scored) AS Total_Runs,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        SUM(b.Runs_Scored) * 1.0 / COUNT(DISTINCT b.Match_Id) AS Avg_Runs_Overall
    FROM ball_by_ball b
    JOIN player p ON b.Striker = p.Player_Id
    GROUP BY p.Player_Id, p.Player_Name
)
SELECT pvs.Player_Id, pvs.Player_Name, pvs.Venue_Name, pvs.Avg_Runs_Venue, pos.Avg_Runs_Overall,
        (pvs.Avg_Runs_Venue - pos.Avg_Runs_Overall) AS Avg_Difference
FROM player_venue_stats pvs
JOIN player_overall_stats pos ON pvs.Player_Id = pos.Player_Id
WHERE pvs.Innings_Played >= 5 -- Consider venues where the player has played at least 5 innings
ORDER BY pvs.Player_Name, Avg_Difference DESC;
```

OUTPUT :

Player_Id	Player_Name	Venue_Name	Avg_Runs_Venue	Avg_Runs_Overall	Avg_Difference
325	A Ashish Reddy	Rajiv Gandhi International Stadium, Uppal	8.87500	13.61111	-4.73611
110	AB de Villiers	M Chinnaswamy Stadium	35.81481	34.52632	1.28849
53	AC Gilchrist	Punjab Cricket Association Stadium, Mohali	17.60000	22.61538	-5.01538
191	AD Mathews	Subrata Roy Sahara Stadium	18.14286	15.80000	2.34286
334	AD Russell	Eden Gardens	27.55556	22.91304	4.64252
254	AJ Finch	Rajiv Gandhi International Stadium, Uppal	31.60000	28.11905	3.48095
254	AJ Finch	Subrata Roy Sahara Stadium	25.28571	28.11905	-2.83334
48	AM Nayar	Subrata Roy Sahara Stadium	7.80000	8.38462	-0.58462
85	AM Rahane	Sardar Patel Stadium, Motera	44.00000	32.40351	11.59649
85	AM Rahane	Maharashtra Cricket Association Stadium	37.66667	32.40351	5.26316

VISUALIZATION :





Player_Id	Player_Name	Venue_Name	Avg_Runs_Venue	Avg_Runs_Overall	Avg_Difference
320	DA Miller	Punjab Cricket A	34	25.59259	8.40741
320	DA Miller	M Chinnaswamy	15.8	25.59259	-9.79259
320	DA Miller	Punjab Cricket A	12.71429	25.59259	-12.8783
187	DA Warner	M Chinnaswamy	52	38.4918	13.5082
187	DA Warner	Rajiv Gandhi Int	45.25	38.4918	6.7582
187	DA Warner	Dr. Y.S. Rajasek	39.8	38.4918	1.3082
187	DA Warner	Feroz Shah Kotl	29.11111	38.4918	-9.38069

2. **Bowlers whose performance are more suited to specific Venues:**

This analysis focuses on the comparison between a bowler's performance at a specific venue and their overall career performance.

APPROACH :

- Calculate `Avg_Wickets_Venue` = Total wickets / Innings played (only for venues with 5 or more innings).
- Calculate `Avg_Wickets_Overall` = Total wickets across all innings / Total innings played.
- Find `Avg_Difference` = `Avg_Wickets_Venue` – `Avg_Wickets_Overall` to measure venue suitability.

INSIGHTS :

- AR Patel at Punjab Cricket Association Stadium shows the highest positive difference of +0.4667, indicating he performs best there.

- A Nehra at Rajiv Gandhi International Stadium records the lowest value of −0.9655, showing clear underperformance.
- Other strong venue specialists include A Nehra at MA Chidambaram Stadium (+0.4345) and DS Kulkarni at Wankhede Stadium (+0.4260).

RECOMMENDATIONS :

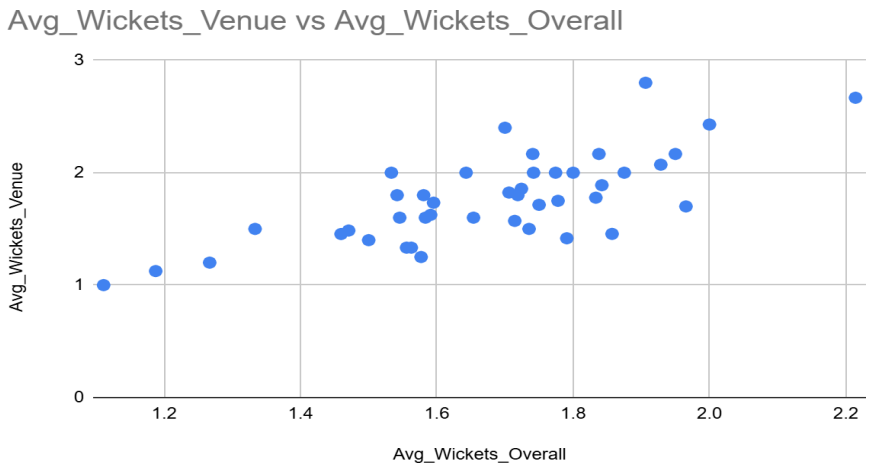
- Select bowlers like AR Patel and DS Kulkarni when matches are held at their suited venues.
- Avoid using bowlers with negative `Avg_Difference`, such as A Nehra at RGI Stadium, for those specific venues.
- Use this data strategically to attack bowlers who underperform at certain venues and plan matchups effectively.

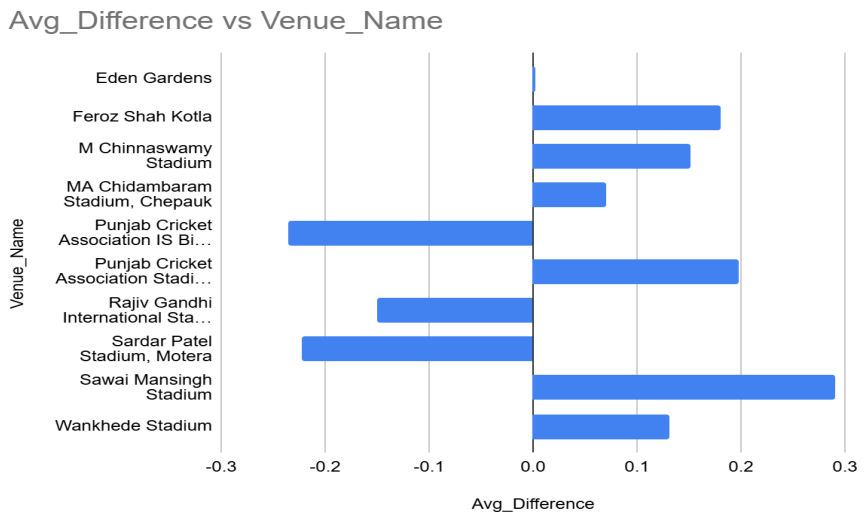
```sql
WITH player_venue_stats AS (
    SELECT p.Player_Id, p.Player_Name, v.Venue_Id, v.Venue_Name,
        COUNT(w.Player_Out) AS Total_Wickets,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        COUNT(w.Player_Out) / COUNT(DISTINCT b.Match_Id) AS Avg_Wickets_Venue
    FROM ball_by_ball b
    JOIN player p ON b.Bowler = p.Player_Id
    JOIN wicket_taken w on (b.Match_Id = w.Match_Id and b.Over_Id = w.Over_Id
                            and b.Ball_Id = w.Ball_Id and b.Innings_No = w.Innings_No)
    JOIN matches m ON b.Match_Id = m.Match_Id
    JOIN venue v ON m.Venue_Id = v.Venue_Id
    GROUP BY p.Player_Id, p.Player_Name, v.Venue_Id, v.Venue_Name
),
player_overall_stats AS (
    SELECT p.Player_Id, p.Player_Name,
        COUNT(w.Player_Out) AS Total_Wickets,
        COUNT(DISTINCT b.Match_Id) AS Innings_Played,
        COUNT(w.Player_Out) / COUNT(DISTINCT b.Match_Id) AS Avg_Wickets_Overall
    FROM ball_by_ball b
    JOIN player p ON b.Bowler = p.Player_Id
    JOIN wicket_taken w on (b.Match_Id = w.Match_Id and b.Over_Id = w.Over_Id
                            and b.Ball_Id = w.Ball_Id and b.Innings_No = w.Innings_No)
    GROUP BY p.Player_Id, p.Player_Name
)
SELECT pvs.Player_Id, pvs.Player_Name, pvs.Venue_Name, pvs.Avg_Wickets_Venue, pos.Avg_Wickets_Overall,
       (pvs.Avg_Wickets_Venue - pos.Avg_Wickets_Overall) AS Avg_Difference
FROM player_venue_stats pvs
JOIN player_overall_stats pos ON pvs.Player_Id = pos.Player_Id
WHERE pvs.Innings_Played >= 5 -- Consider venues where the player has played at least 5 innings
ORDER BY pvs.Player_Name, Avg_Difference DESC;
```

OUTPUT :

Player_Id	Player_Name	Venue_Name	Avg_Wickets_Venue	Avg_Wickets_Overall	Avg_Difference
136	A Mishra	Rajiv Gandhi International Stadium, Uppal	1.9000	1.7188	0.1812
136	A Mishra	Feroz Shah Kotla	1.7000	1.7188	-0.0188
73	A Nehra	MA Chidambaram Stadium, Chepauk	2.4000	1.9655	0.4345
73	A Nehra	Rajiv Gandhi International Stadium, Uppal	1.0000	1.9655	-0.9655
334	AD Russell	Eden Gardens	1.8889	1.8421	0.0468
374	AR Patel	Punjab Cricket Association Stadium, Mohali	2.0000	1.5333	0.4667
299	B Kumar	Rajiv Gandhi International Stadium, Uppal	1.4167	1.7907	-0.3740
350	CH Morris	Feroz Shah Kotla	1.5714	1.7143	-0.1429
71	DJ Bravo	MA Chidambaram Stadium, Chepauk	2.0714	1.9286	0.1428
80	DS Kulkarni	Wankhede Stadium	2.1667	1.7407	0.4260
94	DW Steyn	Rajiv Gandhi International Stadium, Uppal	1.6250	1.5909	0.0341
316	GB Hogg	Eden Gardens	2.0000	1.8750	0.1250
50	Harbhajan S...	Eden Gardens	1.8000	1.5952	0.2048

VISUALIZATION :





Player_Id	Player_Name	Venue_Name	Avg_Wickets_Venue	Avg_Wickets_Overall	Avg_Difference
136	A Mishra	Rajiv Gandhi Int	1.9	1.7188	0.1812
136	A Mishra	Feroz Shah Kotl	1.7	1.7188	-0.0188
73	A Nehra	MA Chidambara	2.4	1.9655	0.4345
73	A Nehra	Rajiv Gandhi Int	1	1.9655	-0.9655